

EMBRYOLOGY PRACTICE QUESTION

1. When does the lower respiratory tract begin to form?

- A. 2nd week
- B. 3rd week
- C. 4th week
- D. 6th week

Answer: C. The development begins during the 4th week.

2. The lower respiratory tract develops from which primordium?

- A. Oropharyngeal membrane
- B. Respiratory diverticulum
- C. Hypopharyngeal eminence
- D. Somatic mesoderm

Answer: B. It develops from the respiratory diverticulum, also known as the lung bud.

3. The respiratory diverticulum first appears as a midline outgrowth called the:

- A. Tracheoesophageal septum
- B. Laryngotracheal groove
- C. Primary bronchial bud
- D. Foramen cecum

Answer: B. It begins as the laryngotracheal groove.

4. From which part of the primitive pharynx does the laryngotracheal groove arise?

- A. Dorsal wall of the midgut
- B. Ventral wall of the foregut
- C. Lateral wall of the hindgut
- D. Cranial wall of the stomodeum

Answer: B. It arises from the caudal part of the ventral wall of the primitive pharynx (foregut).

5. The laryngotracheal groove is located just caudal to which structure?

- A. 1st pharyngeal arch
- B. 2nd pharyngeal pouch
- C. 4th pharyngeal pouch

D. Thyroid gland

Answer: C. It is just caudal to the 4th pharyngeal pouch.

6. What process forms the respiratory diverticulum from the groove?

A. Invagination

B. Evagination

C. Canalization

D. Segmentation

Answer: B. The groove evaginates to form the diverticulum.

7. Initially, the respiratory diverticulum is in open communication with the:

A. Pericardial cavity

B. Primitive pharynx/Foregut

C. Amniotic sac

D. Neural tube

Answer: B. It is initially in open communication with the foregut.

8. Which structures fuse to separate the diverticulum from the foregut?

A. Pleuroperitoneal folds

B. Tracheoesophageal ridges

C. Arytenoid swellings

D. Lingual swellings

Answer: B. Tracheoesophageal ridges (folds) separate them.

9. The fusion of tracheoesophageal ridges forms the:

A. Laryngeal inlet

B. Laryngotracheal tube

C. Tracheoesophageal septum

D. Secondary bronchi

Answer: C. They fuse to form the tracheoesophageal septum.

10. The tracheoesophageal septum divides the foregut into which two portions?

A. Cranial and Caudal

B. Dorsal and Ventral

C. Left and Right

D. Medial and Lateral

Answer: B. It divides it into dorsal and ventral portions.

11. The dorsal portion of the divided foregut becomes the primordium of the:

- A. Trachea and Lungs
- B. Larynx
- C. Oropharynx and Esophagus
- D. Heart

Answer: C. It becomes the oropharynx and esophagus.

12. The ventral portion of the divided foregut is known as the:

- A. Lung bud
- B. Laryngotracheal tube
- C. Primary bronchus
- D. Pharyngeal pouch

Answer: B. It is the laryngotracheal tube.

13. The laryngotracheal tube is the primordium for which structures?

- A. Nose and Pharynx
- B. Larynx, Trachea, Bronchi, and Lungs
- C. Esophagus and Stomach
- D. Only the Alveoli

Answer: B. It forms the entire lower respiratory tract.

14. The epithelium and glands of the respiratory tract are derived from:

- A. Ectoderm
- B. Mesoderm
- C. Endoderm
- D. Neural Crest

Answer: C. They are derived from the endoderm lining the diverticulum.

15. The connective tissue, cartilage, and smooth muscle of the respiratory tract are derived from:

- A. Somatic mesoderm
- B. Splanchnic mesoderm
- C. Surface ectoderm
- D. Endoderm

Answer: B. They are derived from the surrounding splanchnic mesoderm.

16. The splanchnic mesoderm involved in respiratory development is derived from:

- A. Yolk sac
- B. Neural crest cells
- C. Trophoblast
- D. Paraxial mesoderm

Answer: B. It is derived from neural crest cells.

17. Which of the following is NOT part of the lower respiratory tract?

- A. Larynx
- B. Trachea
- C. Pharynx
- D. Lungs

Answer: C. The pharynx is part of the upper respiratory tract.

18. The proximal part of the respiratory diverticulum remains tubular and forms the:

- A. Alveoli
- B. Larynx and Trachea
- C. Primary bronchi
- D. Pleural cavity

Answer: B. It forms the larynx and trachea.

19. The distal end of the respiratory diverticulum dilates to form the:

- A. Laryngeal inlet
- B. Epiglottis
- C. Lung bud
- D. Esophagus

Answer: C. It forms the lung bud.

20. The communication between the respiratory primordium and pharynx persists as the:

- A. Foramen cecum
- B. Laryngeal inlet
- C. Vocal cord
- D. Tracheoesophageal fistula

Answer: B. It persists as the laryngeal inlet.

Part 2: Development of the Larynx and Epiglottis (Questions 21–40)

21. The larynx develops from which part of the laryngotracheal tube?

- A. Distal part
- B. Cranial-most part
- C. Middle part
- D. Bifurcation point

Answer: B. It develops from the cranial-most part.

22. The laryngeal muscles and cartilages (except the epiglottis) develop from which pharyngeal arches?

- A. 1st and 2nd
- B. 2nd and 3rd
- C. 4th and 6th
- D. 5th and 6th

Answer: C. They develop from the 4th and 6th arches.

23. Which cells give rise to the mesoderm of the 4th and 6th pharyngeal arches?

- A. Endodermal cells
- B. Neural crest cells
- C. Ectodermal cells
- D. Somites

Answer: B. They are derived from neural crest cells.

24. Proliferation of mesenchyme surrounding the laryngeal orifice causes it to become:

- A. Circular
- B. T-shaped
- C. Square
- D. O-shaped

Answer: B. It becomes T-shaped.

25. Which of the following is NOT a cartilage formed from the 4th and 6th arch mesenchyme?

- A. Thyroid
- B. Cricoid
- C. Arytenoid
- D. Epiglottis

Answer: D. The epiglottis develops differently.

26. The epiglottis develops from the caudal part of the:

- A. Laryngotracheal groove
- B. Hypopharyngeal eminence
- C. Primary bronchial bud
- D. Tracheoesophageal septum

Answer: B. It develops from the hypopharyngeal eminence.

27. The hypopharyngeal eminence is formed by mesoderm in the ventral ends of which arches?

- A. 1st and 2nd
- B. 3rd and 4th
- C. 4th and 6th
- D. 2nd and 3rd

Answer: B. It is formed from the 3rd and 4th arches.

28. At what age does the epiglottis reach its adult form?

- A. At birth
- B. 1 year
- C. 3 years
- D. 8 years

Answer: C. It reaches adult form by the first three years after birth.

29. What causes the temporary occlusion of the laryngeal lumen?

- A. Cartilage growth
- B. Rapid epithelial proliferation
- C. Fluid accumulation
- D. Muscle contraction

Answer: B. It is caused by rapid proliferation of laryngeal epithelium.

30. When does recanalization of the larynx typically take place?

- A. 4th week
- B. 7th week
- C. 10th week
- D. 24th week

Answer: C. It takes place by the 10th week.

31. During recanalization, the upper pair of endodermal folds forms the:

- A. True vocal cords
- B. Vestibular folds (false vocal cords)
- C. Alveolar ducts
- D. Arytenoid swellings

Answer: B. They form the vestibular folds.

32. The lower pair of endodermal folds during recanalization forms the:

- A. False vocal cords
- B. Vocal folds (true vocal cords)
- C. Laryngeal ventricles
- D. Epiglottis

Answer: B. They form the vocal folds.

33. The lateral recesses bounded by the vestibular and vocal folds are called:

- A. Bronchioles
- B. Pleural cavities
- C. Laryngeal ventricles
- D. Sinuses

Answer: C. They are called laryngeal ventricles.

34. Failure of laryngeal recanalization leads to:

- A. Tracheoesophageal fistula
- B. Laryngeal atresia
- C. Lung hypoplasia
- D. Esophageal atresia

Answer: B. It leads to laryngeal atresia.

35. The arytenoid swellings are visible during which week of development?

- A. 4th week
- B. 6th week (based on diagrams)
- C. 10th week
- D. 24th week

Answer: B. Diagram B in slide 13 shows these swellings early in development.

36. Which germ layer forms the laryngeal glands?

- A. Splanchnic mesoderm

- B. Endoderm
- C. Somatic mesoderm
- D. Neural crest

Answer: B. Glands are derived from endoderm.

37. Growth of the larynx and epiglottis is most rapid during:

- A. The first trimester
- B. The third trimester
- C. The first three years after birth
- D. Puberty

Answer: C. Growth is rapid during the first three years after birth.

38. Which structure lies between the foramen cecum and the laryngeal inlet?

- A. Epiglottis
- B. Trachea
- C. Esophagus
- D. Primary bronchus

Answer: A. The epiglottis/hypopharyngeal eminence is between them.

39. The characteristic adult shape of the laryngeal orifice is recognized after:

- A. Formation of the groove
- B. Mesenchyme transforms into thyroid, cricoid, and arytenoid cartilages
- C. Birth
- D. The 24th week

Answer: B. It is recognized when these cartilages form.

40. Laryngeal atresia causes obstruction of the:

- A. Lower fetal airway
- B. Upper fetal airway
- C. Esophagus
- D. Nasal cavity

Answer: B. It produces obstruction of the upper fetal airway.

Part 3: Development of the Trachea (Questions 41–55)

41. The trachea develops from the part of the laryngotracheal tube located:

- A. Cranial to the larynx

- B. Between the larynx and the bifurcation
- C. Distal to the lungs
- D. Inside the pleural cavity

Answer: B. It is the portion between the larynx and bifurcation.

42. The lining epithelium of the trachea is derived from:

- A. Splanchnic mesoderm
- B. Endoderm
- C. Somatic mesoderm
- D. Ectoderm

Answer: B. It comes from endoderm.

43. Tracheal cartilages and smooth muscles are derived from:

- A. Endoderm
- B. Splanchnopleuric mesoderm
- C. Pharyngeal arches 1 and 2
- D. Surface ectoderm

Answer: B. They are derived from the surrounding splanchnopleuric mesoderm.

44. By which week is cartilage already developing around the tracheal lumen?

- A. 4 weeks
- B. 6 weeks
- C. 11 weeks
- D. 24 weeks

Answer: B. It is developing by 6 weeks.

45. By which week are tracheal glands and distinct smooth muscle present?

- A. 6 weeks
- B. 11 weeks
- C. 20 weeks
- D. After birth

Answer: B. They are present by 11 weeks.

46. What is "A" in the 11-week tracheal diagram?

- A. Cartilage
- B. Epithelium

- C. Smooth muscle
- D. Connective tissue

Answer: C. Smooth muscle is identified in the 11-week diagram.

47. Tracheal stenosis refers to:

- A. Complete absence of the trachea
- B. Narrowing of the trachea
- C. An abnormal passage to the esophagus
- D. Extra lung tissue

Answer: B. Stenosis means narrowing.

48. Tracheal atresia refers to:

- A. Narrowing
- B. Blockage or absence of the lumen
- C. Inflammation
- D. Fluid in the lungs

Answer: B. Atresia implies a blockage or failure to form a lumen.

49. Incomplete division of the foregut during the 4th week results in:

- A. Laryngeal atresia
- B. Tracheoesophageal fistula
- C. Lung agenesis
- D. Respiratory distress syndrome

Answer: B. It results in TEF.

50. Tracheoesophageal fistula is generally associated with:

- A. Lung hypoplasia
- B. Esophageal atresia
- C. Accessory lungs
- D. Hyperactive surfactant

Answer: B. It is generally associated with esophageal atresia.

51. Esophageal atresia in a fetus can lead to:

- A. Oligohydramnios
- B. Polyhydramnios
- C. Low birth weight

D. Rapid lung maturation

Answer: B. It results in polyhydramnios.

52. Why does esophageal atresia cause polyhydramnios?

A. The fetus cannot swallow amniotic fluid

B. The lungs produce too much fluid

C. The kidneys fail

D. Surfactant is missing

Answer: A. The fetus cannot swallow amniotic fluid, leading to its accumulation.

53. The diagram showing a fistula connecting the upper esophagus to the trachea represents:

A. Type A TEF

B. Type B TEF

C. A common variation of TEF

D. Normal development

Answer: C. Variations A-D are shown in slide 49.

54. The surrounding mesenchyme of the trachea originates from:

A. Somatic mesoderm

B. Neural crest cells

C. Endoderm

D. Trophoblast

Answer: B. Splanchnic mesoderm is derived from neural crest cells.

55. The tracheal lumen is initially lined by:

A. Squamous cells

B. Endodermal epithelium

C. Smooth muscle

D. Cartilage

Answer: B. It is lined by endoderm-derived epithelium.

Part 4: Bronchi, Lungs, and Pleura (Questions 56–75)

56. The respiratory diverticulum divides into how many bronchial buds?

A. One

B. Two

C. Three

D. Five

Answer: B. It divides into two bronchial buds.

57. Bronchial buds grow laterally into which spaces?

A. Peritoneal cavity

B. Pericardioperitoneal canals

C. Neural canal

D. Oral cavity

Answer: B. They grow into the pericardioperitoneal canals.

58. The pericardioperitoneal canals are primordia of the:

A. Heart

B. Pleural cavities

C. Diaphragm

D. Stomach

Answer: B. They are primordia of the pleural cavities.

59. When does each bronchial bud develop into a principal (main) bronchus?

A. 4th week

B. 5th week

C. 7th week

D. 10th week

Answer: B. This occurs at the beginning of the 5th week.

60. Compared to the left, the right main bronchus is:

A. Smaller and more horizontal

B. Larger and more vertical

C. Longer and thinner

D. Identical

Answer: B. The right is larger and more vertical.

61. Which bronchus lies more transversely?

A. Right main bronchus

B. Left main bronchus

C. Right superior lobar bronchus

D. Trachea

Answer: B. The left main bronchus lies more transversely.

62. The secondary bronchi are also known as:

- A. Principal bronchi
- B. Lobar bronchi
- C. Segmental bronchi
- D. Terminal bronchioles

Answer: B. They are lobar bronchi.

63. On the right side, the inferior secondary bronchus subdivides into:

- A. Two bronchi (middle and lower lobes)
- B. Three bronchi
- C. Only one bronchus
- D. Ten bronchi

Answer: A. It supplies the middle and lower lobes.

64. How many secondary bronchi are in the left lung?

- A. One
- B. Two
- C. Three
- D. Ten

Answer: B. There are two (upper and lower lobes).

65. When do segmental (tertiary) bronchi begin to form?

- A. 5th week
- B. 7th week
- C. 16th week
- D. 24th week

Answer: B. They form by the 7th week.

66. How many segmental bronchi form in the right lung?

- A. 8
- B. 9
- C. 10
- D. 12

Answer: C. There are 10 in the right lung.

67. How many segmental bronchi form in the left lung?

- A. 5 or 6
- B. 8 or 9
- C. 10 or 11
- D. 15

Answer: B. There are 8 or 9 in the left lung.

68. What forms the primordia of the bronchopulmonary segments?

- A. Lobar bronchi and pleura
- B. Segmental bronchi and surrounding mesenchyme
- C. Alveoli and capillaries
- D. Trachea and esophagus

Answer: B. Segmental bronchi plus mesenchyme form them.

69. By 24 weeks, how many orders of branches have typically formed?

- A. 5
- B. 10
- C. 17
- D. 24

Answer: C. Approximately 17 orders have formed.

70. How many additional orders of airways develop after birth?

- A. None
- B. Two
- C. Seven
- D. Seventeen

Answer: C. Seven additional orders develop postnatally.

71. The visceral pleura is derived from:

- A. Endoderm
- B. Splanchnic mesoderm
- C. Somatic mesoderm
- D. Neural crest

Answer: B. It comes from splanchnic mesoderm.

72. The parietal pleura is derived from:

- A. Endoderm
- B. Splanchnic mesoderm
- C. Somatic mesoderm
- D. Ectoderm

Answer: C. It comes from somatic mesoderm.

73. The space between the parietal and visceral pleura is the:

- A. Pericardial cavity
- B. Pleural cavity
- C. Peritoneal cavity
- D. Mediastinum

Answer: B. It is the pleural cavity.

74. Which structure is depicted at 28 days in the branching diagram?

- A. Segmental bronchi
- B. Secondary bronchial buds
- C. Alveoli
- D. Mature lungs

Answer: B. Secondary bronchial buds are shown at 28 days.

75. Which structure is depicted at 56 days?

- A. Lung bud initiation
- B. Segmental (tertiary) bronchi
- C. Respiratory bronchioles
- D. Terminal sacs

Answer: B. Segmental bronchi are shown at 56 days.

Part 5: Maturation of the Lungs (Questions 76–95)

76. Lung maturation is divided into how many periods?

- A. Two
- B. Three
- C. Four
- D. Five

Answer: C. It is divided into four periods.

77. Why do the periods of lung maturation overlap?

- A. Because of different germ layers
- B. Cranial segments mature faster than caudal ones
- C. Because the left lung matures faster than the right
- D. They do not overlap

Answer: B. Cranial segments mature faster.

78. What is the first stage of lung maturation?

- A. Canalicular
- B. Alveolar
- C. Pseudoglandular
- D. Terminal sac

Answer: C. The pseudoglandular stage is first.

79. The pseudoglandular stage occurs during which weeks?

- A. 1–5 weeks
- B. 5–17 weeks
- C. 16–25 weeks
- D. 24 weeks to birth

Answer: B. It occurs from 5 to 17 weeks.

80. In the pseudoglandular stage, the lungs resemble:

- A. Muscle tissue
- B. Exocrine glands
- C. Sponges
- D. Neural tissue

Answer: B. They resemble exocrine glands.

81. By 16 weeks, the bronchial tree is formed up to:

- A. Alveoli
- B. Terminal bronchioles
- C. Respiratory bronchioles
- D. Alveolar ducts

Answer: B. It is formed up to terminal bronchioles.

82. Is respiration possible during the pseudoglandular stage?

- A. Yes, always

- B. Only at 17 weeks
- C. No
- D. Only if given surfactant

Answer: C. Respiration is NOT possible.

83. The canalicular stage occurs during which weeks?

- A. 5–17 weeks
- B. 16–25 weeks
- C. 24 weeks to birth
- D. 32 weeks to 8 years

Answer: B. It occurs from 16 to 25 weeks.

84. What develops from terminal bronchioles during the canalicular stage?

- A. Segmental bronchi
- B. Respiratory bronchioles
- C. Tracheal glands
- D. Vocal cords

Answer: B. They give rise to respiratory bronchioles.

85. Respiratory bronchioles divide into 3 to 6 tubular passages called:

- A. Alveoli
- B. Alveolar ducts
- C. Lobar bronchi
- D. Terminal sacs

Answer: B. They divide into alveolar ducts.

86. Respiration becomes possible at the end of which stage?

- A. Pseudoglandular
- B. Canalicular
- C. Terminal sac
- D. Alveolar

Answer: B. It is possible at the end of the canalicular stage.

87. The terminal sac period occurs from:

- A. 5–17 weeks
- B. 16–25 weeks

- C. 24 weeks to birth
- D. Late fetal period to 8 years

Answer: C. It is from 24 weeks to birth.

88. What are primordial alveoli called?

- A. Alveolar ducts
- B. Terminal sacs
- C. Bronchioles
- D. Pneumocytes

Answer: B. They are terminal sacs.

89. By 24 weeks, terminal sacs are lined by which cells?

- A. Type I and Type II pneumocytes
- B. Ciliated cells
- C. Goblet cells
- D. Smooth muscle cells

Answer: A. They are lined by Type I and Type II pneumocytes.

90. Which cells are responsible for gaseous exchange?

- A. Type II pneumocytes
- B. Type I pneumocytes
- C. Macrophages
- D. Endothelial cells

Answer: B. Squamous Type I pneumocytes perform gas exchange.

91. What do Type II pneumocytes secrete?

- A. Mucus
- B. Surfactant
- C. Hormones
- D. Plasma

Answer: B. They secrete surfactant.

92. Surfactant's primary function is to:

- A. Carry oxygen
- B. Lower surface tension
- C. Kill bacteria

D. Trap dust

Answer: B. It lowers surface tension at the air-alveolar interface.

93. The blood-air barrier is formed by the contact of:

- A. Alveolar epithelium and capillary endothelium
- B. Cartilage and muscle
- C. Type I and Type II cells
- D. Pleura and lung tissue

Answer: A. It is the contact between epithelial and endothelial cells.

94. Adequate surfactant for survival is typically produced during the:

- A. Pseudoglandular stage
- B. Last 2 weeks before birth
- C. First year after birth
- D. Canalicular stage

Answer: B. Sufficient amounts are produced in the last 2 weeks before birth.

95. The alveolar stage lasts from:

- A. 24 weeks to birth
- B. Late fetal period to 8 years
- C. Birth to 3 years
- D. 11 weeks to 20 weeks

Answer: B. It lasts until approximately 8 years of age.

Part 6: Postnatal Development and Clinical (Questions 96–100)

96. What percentage of mature alveoli develop after birth?

- A. 5%
- B. 50%
- C. 75%
- D. 95%

Answer: D. 95% of mature alveoli develop postnatally.

97. How many alveoli are present at birth?

- A. 50 million
- B. 150 million
- C. 300 million

D. 500 million

Answer: A. About 50 million (one-sixth of the adult number) are present.

98. How many alveoli are present in an adult (by age 8)?

A. 50 million

B. 100 million

C. 300 million

D. 1 billion

Answer: C. The adult complement is 300 million.

99. Lungs increase in size primarily due to:

A. Increase in size of existing alveoli

B. Increase in the number of respiratory bronchioles and alveoli

C. Accumulation of fluid

D. Growth of the rib cage

Answer: B. It is due to an increase in number, not size.

100. Which of the following is an anomaly of the LRT?

A. Lung hypoplasia

B. Agenesis of lungs

C. Congenital lung cysts

D. All of the above

Answer: D. All listed are anomalies.

